



CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 08



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Recap of Previous Lecture



Topic

Types of Graph Part-04



Topics to be Covered



Topic

Definition in Connectivity

Topic

Connected vs Disconnected

Topic

Range of Edges

Topic

Concepts of Tree

Topic

Connectivity Theorem



Topic : Connectivity in Graphs



Walk:

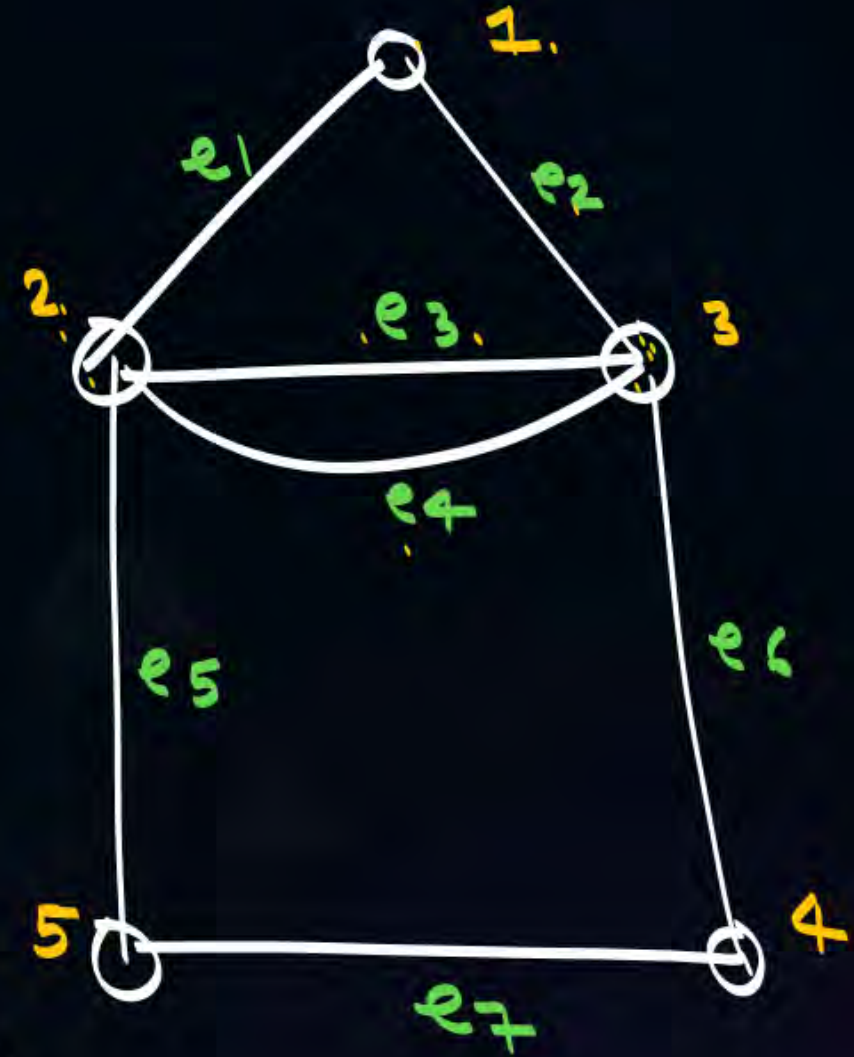
alternating sequences of vertices & edges.
1 e_2 3 e_3 2 e_3 3 e_6 4 $R.V | R.E$

Trail:

alternating sequences of vertices & edges.
1 e_2 ③ e_3 2 e_4 ③ e_6 4 $R.V | R.E$

Path:

alternating sequences of vertices & edges.
1 e_2 3 e_6 4 $R.V | R.E$





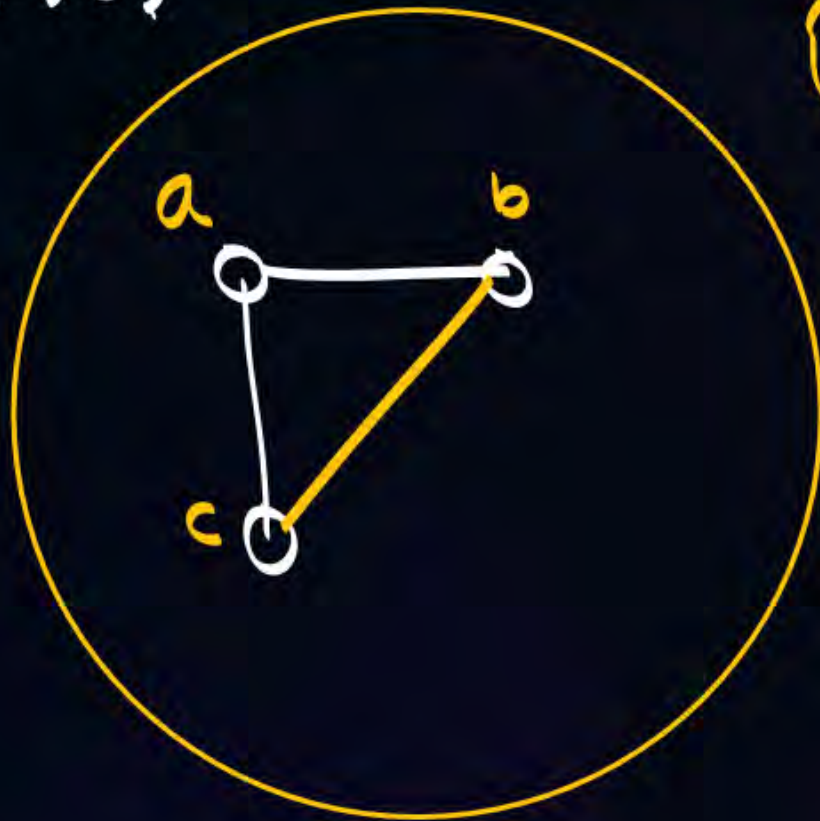
Topic : Connectivity in Graphs

Graph

($k=1$)

Connected:

$G=(V, E)$



path is available.
betⁿ all pair of
vertices.

if path is not
available for atleast
1 pair, then it
is disconnected.
graph.

($k \geq 2$)

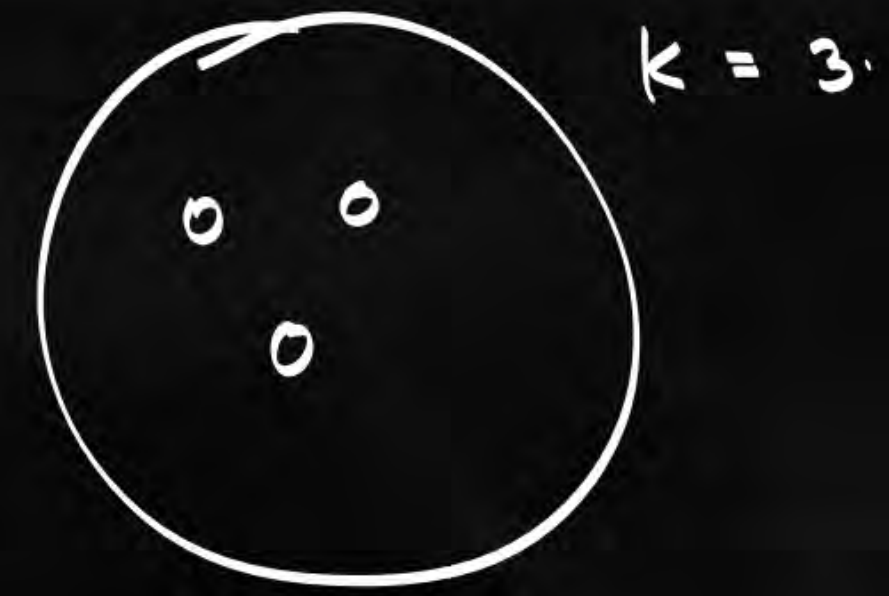
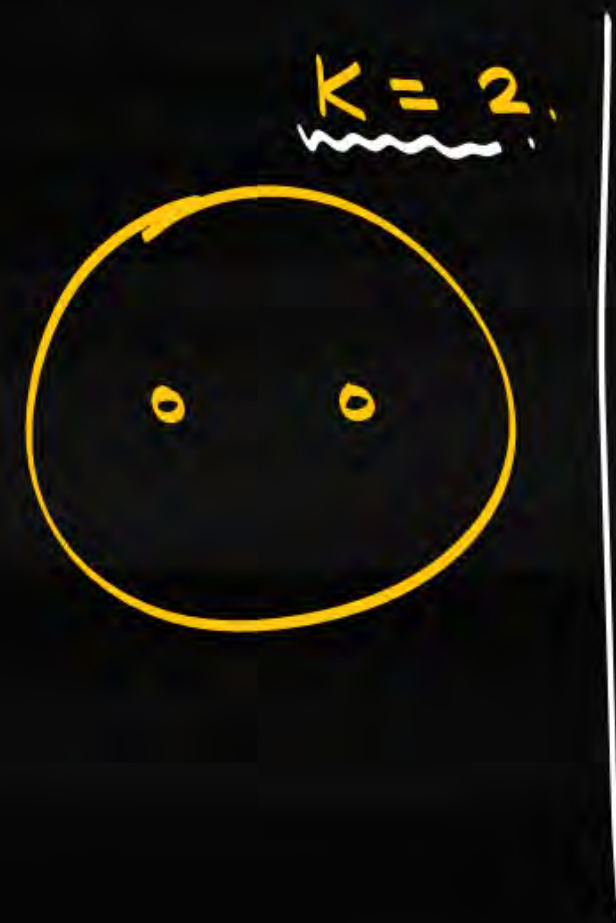
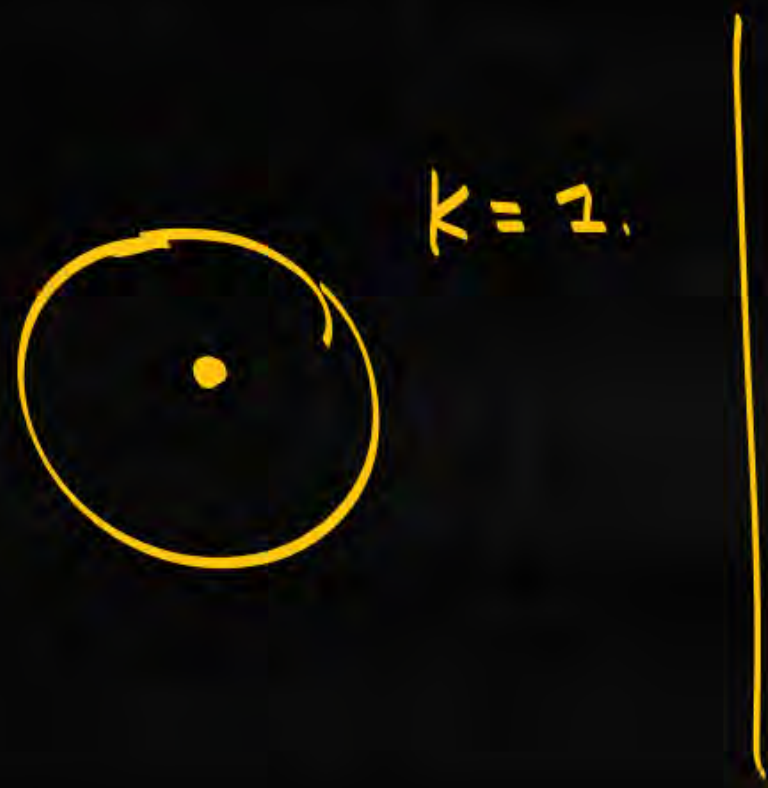
Disconnected:

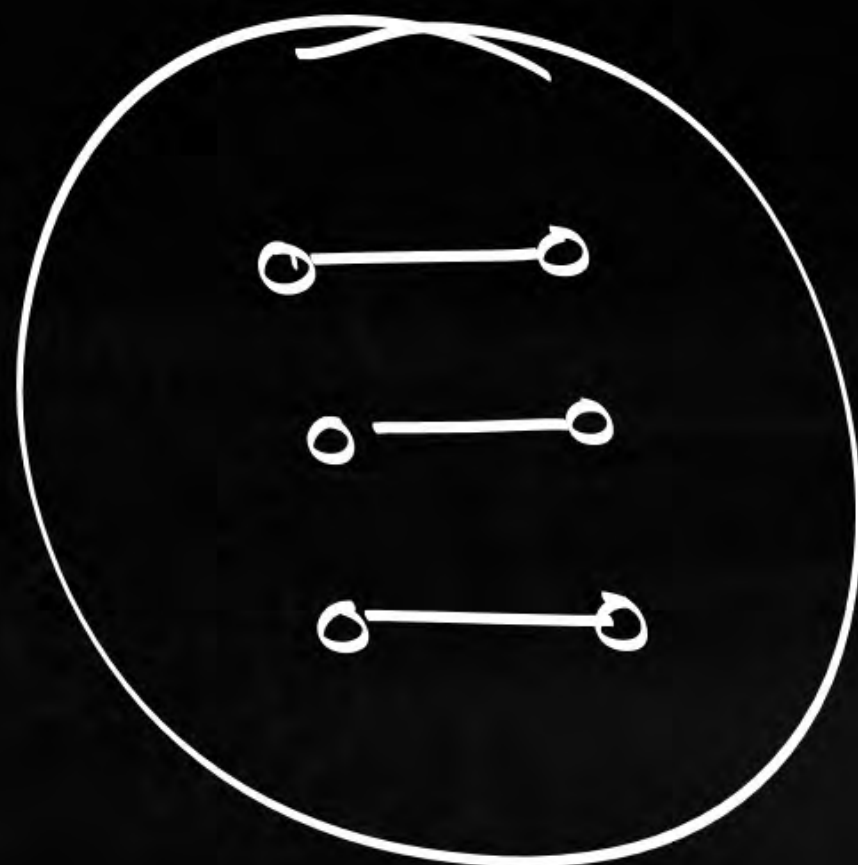
$G=(V, E)$

$|V|=7$



Disconnected Graph contains connected subparts
Component(K)





$K=3$:

Q: Consider a star Graph, of 6 vertices

6 vertices star Graph.

$K_{1,5}$



o



when we take complement of star Graph?

a) no. of edges in Resultant Graph.

b) no. of components.

$$e(G) + e(\bar{G}) = \frac{n(n-1)}{2} \quad (n=6)$$

$$\downarrow$$

$$5 + x = \frac{6 \cdot 5}{2} = 15$$

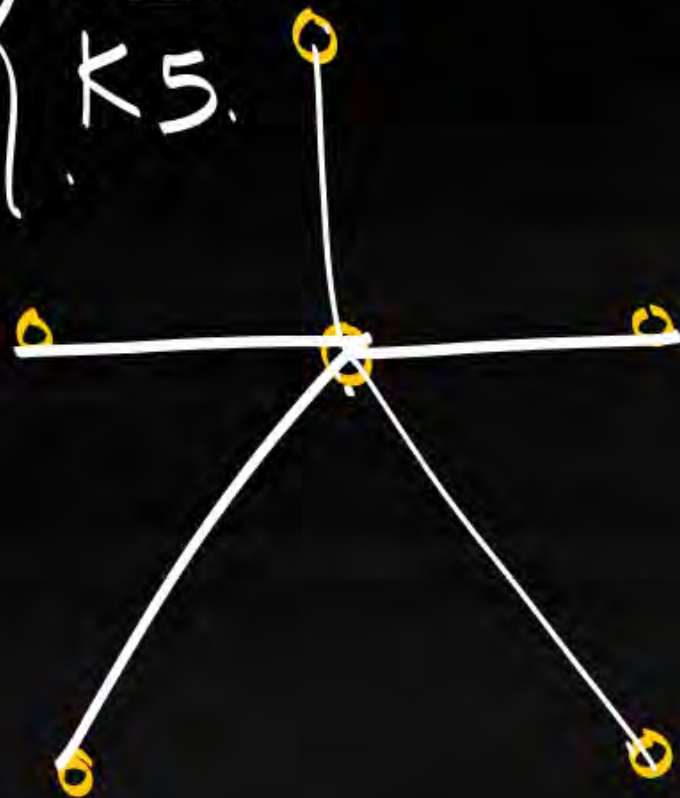
$$x = 10$$

$K_{1,5}$



OR

$\begin{cases} K_1 \\ K_5 \end{cases}$



Star Graph.

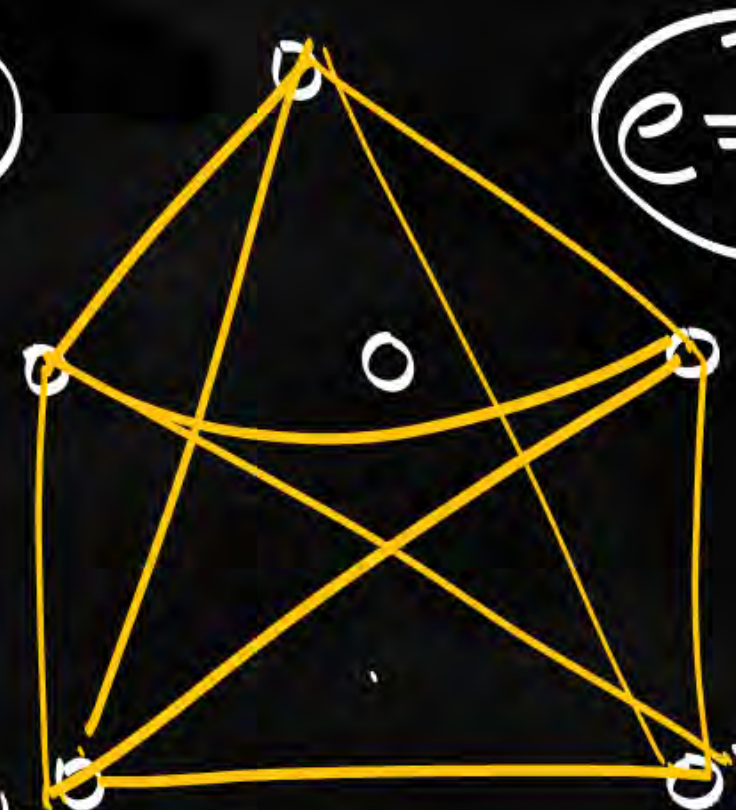
$K_{1, n-1}$

K_2 K_{n-1}

Complement

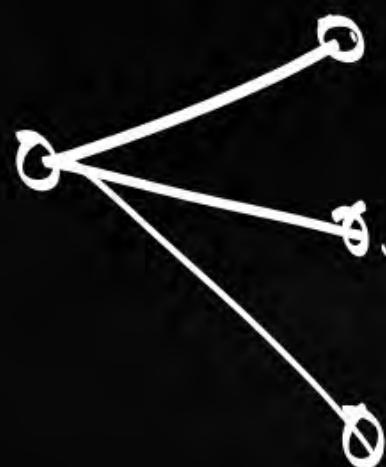


2 component
 $k=2$ $\begin{cases} K_1 \\ K_5 \end{cases}$



$e=10$

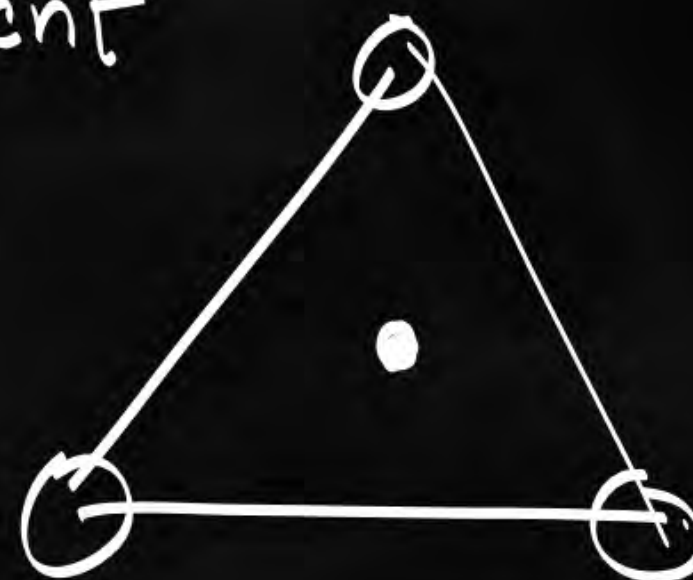
$K_{1,3}$



OR



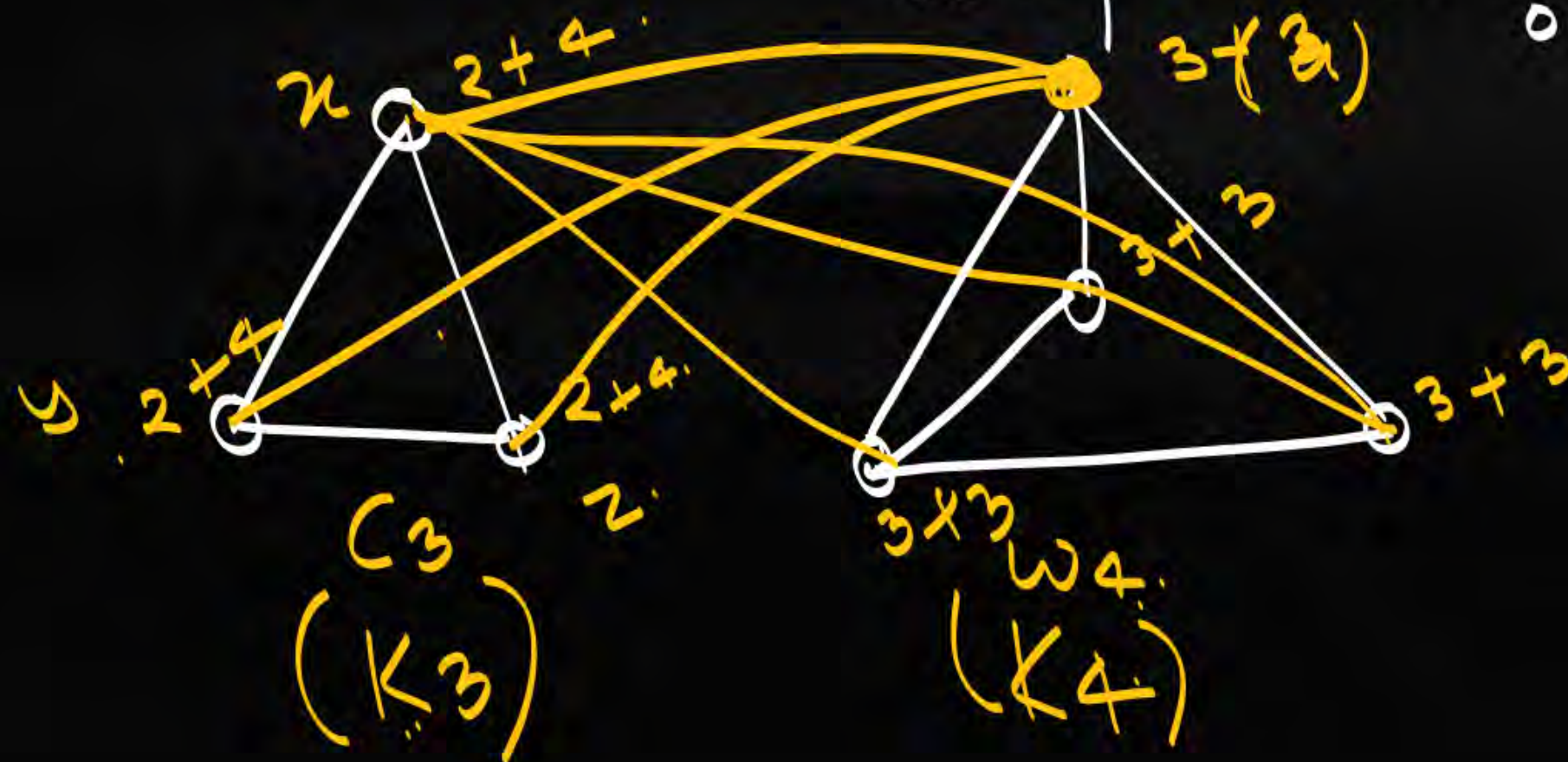
Complement



if we connect all vertices C_3 to all vertices of W_4 .
we will get resultant Graph.

if we take complement of resultant Graph.
what will be no. of

components?



Ans:
 $K=7$

7 vertices.
Degree of each vertex is 6.
 $\rightarrow K_7$.

$$e(\overline{K_7}) = 0$$

$$\underline{\underline{K_3}} * \underline{\underline{K_4}} = K_7.$$

$$K_m * K_n = K_{m+n}.$$

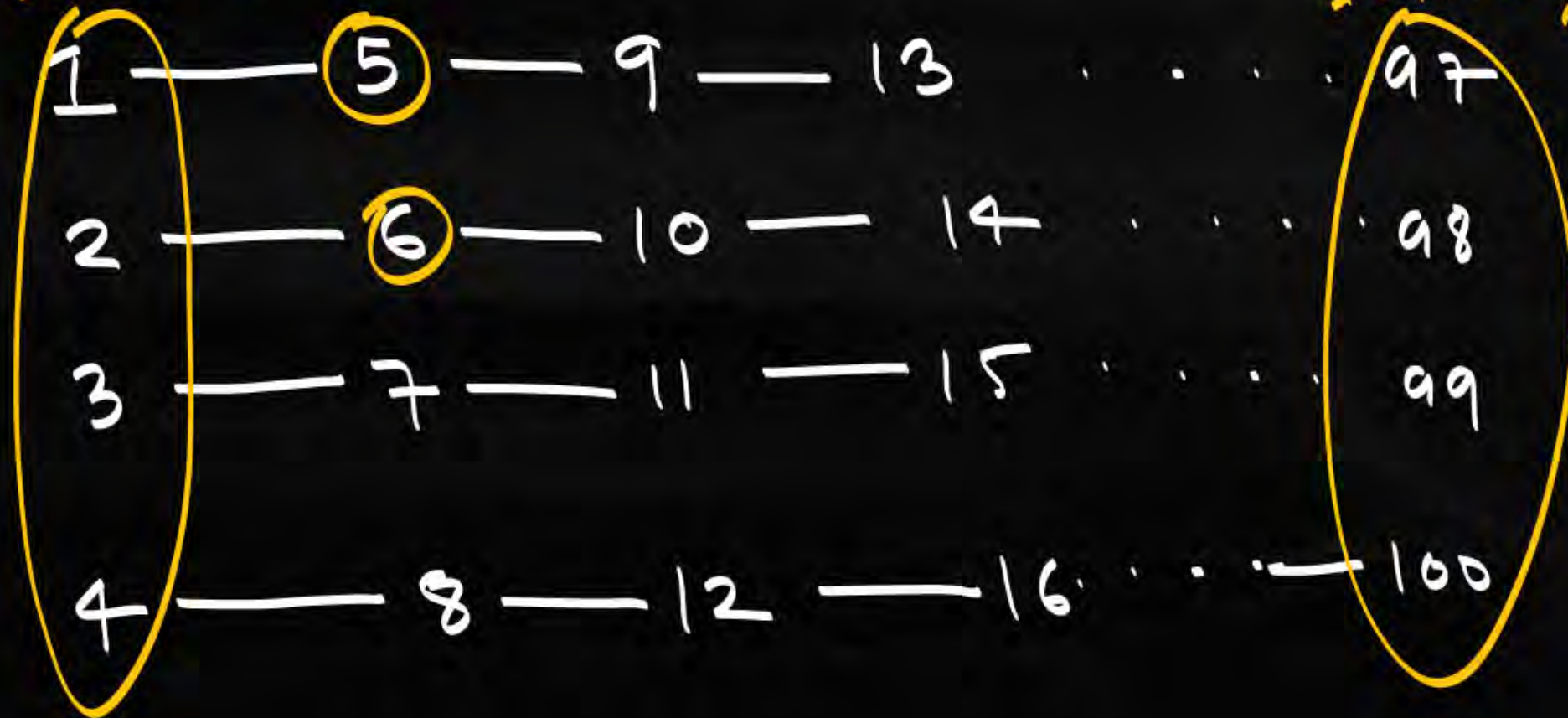
Connect
all vertices of K_m
with all vertices of K_n .

* Consider a Graph vertices are represented as no. from $\{1 \dots 100\}$, two vertices will be connected (x, y)

Degree 2

$$|x - y| = 4$$

1) what will be no. of edges? b) $k = ?$



$$k = 4$$

$$\sum d(v_i) = 2e$$

$$4 \times 1 + 4 \times 1$$

$$+ 92 \times 2 = 2e$$

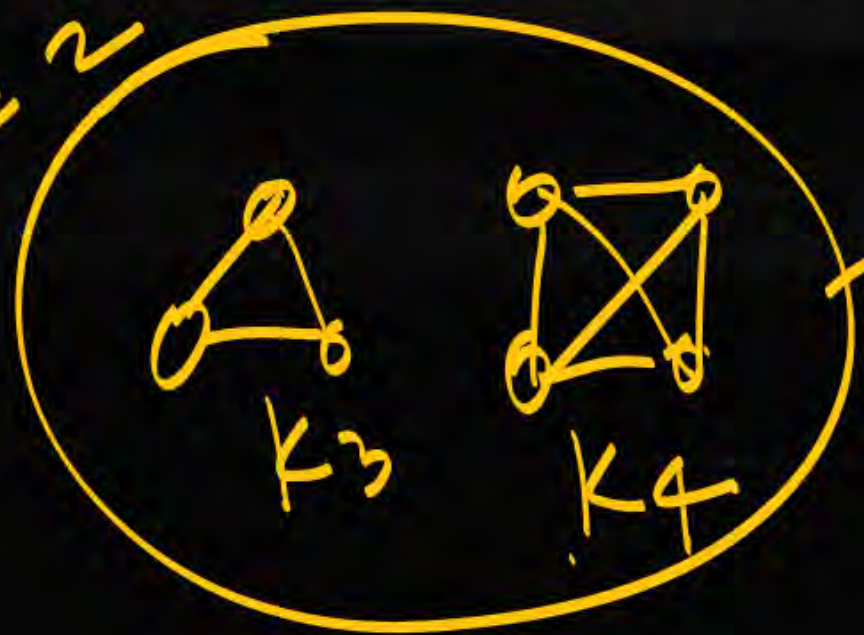
$$e = 96$$

S1: if G is connected, then $\bar{G}$ will be connected? (false)



S2: if G is disconnected then $\bar{G}$ will be connected? (true)

$K=2$



Q: Consider a Graph of 10 vertices, $\delta(G) \geq 5$

check it is connected ✓

Let's assume
it is Disconnected Graph.

$n=10$ $\delta(G) \geq 5$

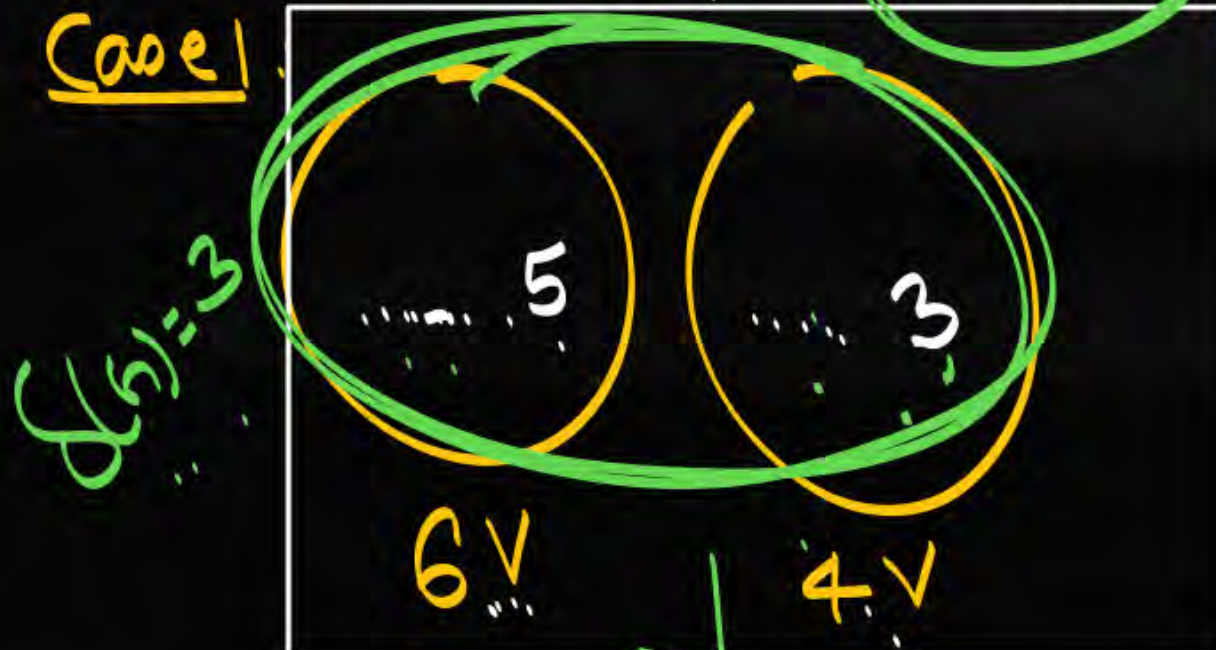
OR
Disconnected?

let's take $k=2$

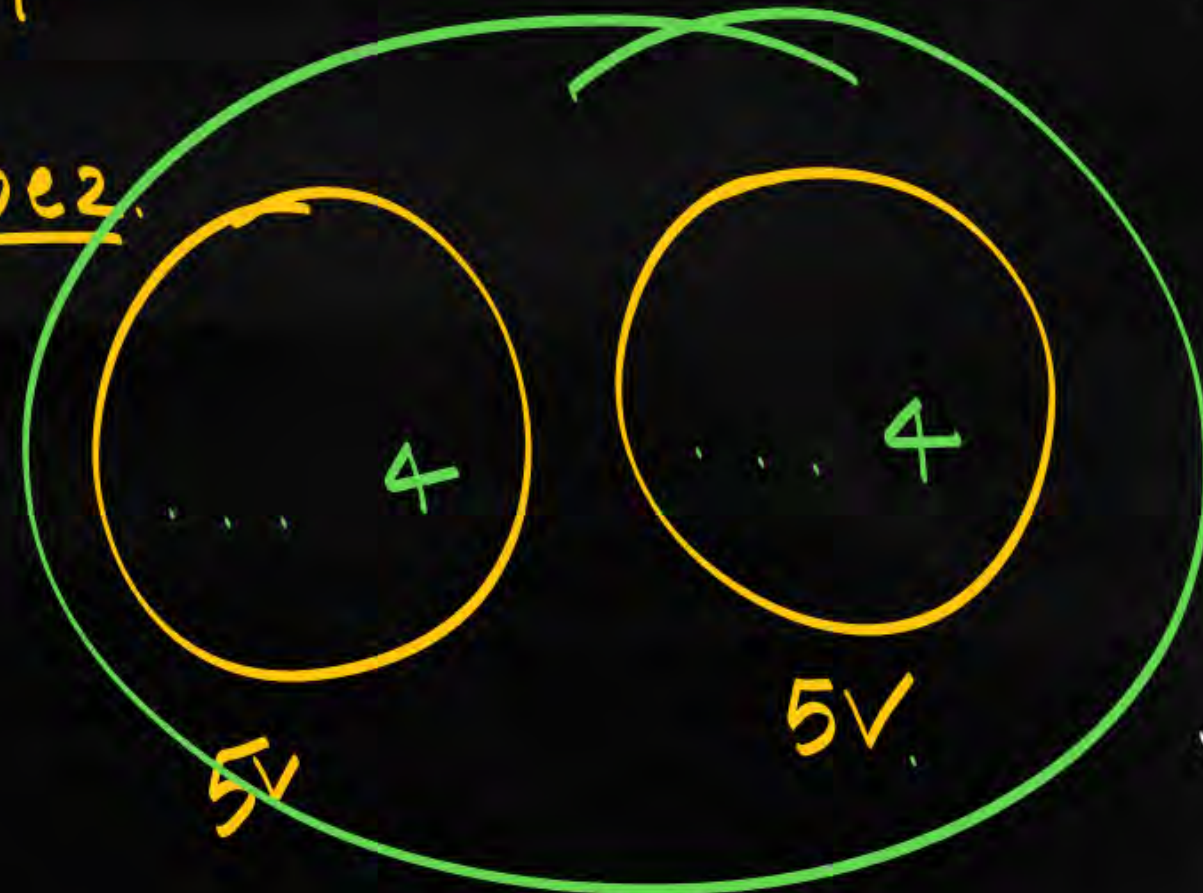
$n=10$

$\delta(G) \geq 5$

Case 1.



Case 2.



$\delta(G)=4$

Thm: if $\delta(G) \geq \frac{n-1}{2}$ then it is connected.

but vice versa is not true.

eg: C_7 .

vice versa part

if G is connected then $\delta(G) \geq \frac{n-1}{2}$ (false)

eg: C_7

$n=7$

$\delta(G) \geq \frac{7-1}{2}$

$\delta(G) \geq 3$

Thm.:

Consider a Graph is having exactly 2 odd degree vertices.
 then, path is available between those.
 2 vertices (True)

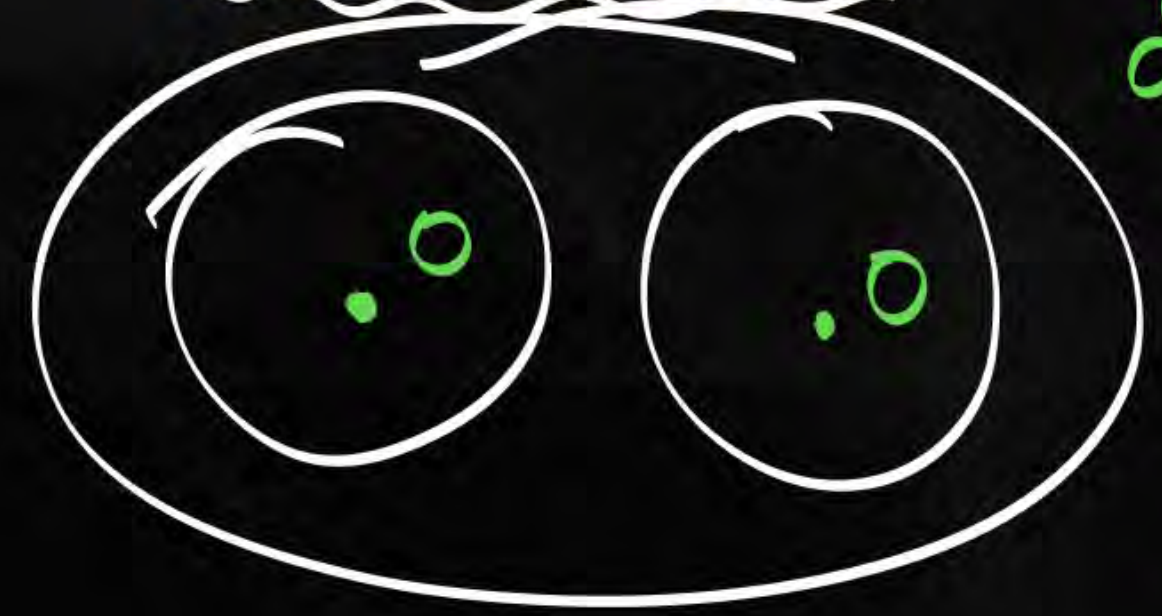
Graph.

Connected.

↓
 path is available
 betn all.

↓
 true.

Disconnected.



exactly
 2 odd
 degree
 vertices.

Thm 2



2 mins Summary

Topic

One

Definition in Connectivity

Topic

Two

Connected vs Disconnected

Topic

Three

Range of Edges

Topic

Four

Topic

Five



THANK - YOU